

Damodar Dasi

Senior Mechanical & Package Design Engineer with experience across compressor packages, PSA nitrogen systems, fuel systems, storage tanks, mechanical skids and integrated package design.

PROFILE

Mechanical & Package Design

EXPERIENCE

6+ years overall · 4+ compressor/package

LOCATION

Sharjah, UAE

CONTACT

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LinkedIn profile

CORE SOFTWARE

SolidWorks · AutoCAD

PROFESSIONAL SUMMARY

I work at the intersection of mechanical design, package integration and project execution. My experience includes complete 3D package development, equipment arrangement, process and utility piping, lubrication systems, seal systems, structural/base frames, engineering drawings, MTO/BOM, multidisciplinary coordination, manufacturing support and FAT/SAT activities.

My project exposure includes international compressor packages, an ONGC PSA nitrogen generator, an Emarat mobile fueling unit, RFQ-stage fuel systems, storage tanks and mechanical/structural skids.

PROFESSIONAL EXPERIENCE

Nov 2025 – Present

Mechanical Design & Project Engineer

Meta Green Middle East FZ LLC · UAE

- Mechanical and package design for mobile and skid-mounted fueling systems, equipment packages, storage tanks and fabricated equipment.
- 3D models, GA/fabrication drawings and BOM/BOQ/MTO in SolidWorks and AutoCAD.
- Skid structures, equipment supports, piping, tanks, pumps and instrumentation; structural/mechanical calculations including weight, CoG and lifting considerations.
- Head of Design Department, Project Coordinator and Lead Mechanical Designer for the Emarat 40 FT dual-compartment MFU — tender through design, procurement, manufacturing, FAT and SAT.
- Approximately 25 storage tank designs from 500 L to 60 KL.

Apr 2023 – Oct 2025

Senior Design Engineer — Compressor, Compander & PSA Nitrogen Packages

DesignTech Systems · Atlas Copco onsite

- Led mechanical and package design for centrifugal compressor/compander packages and PSA nitrogen generator systems.
- Developed 3D assemblies in SolidWorks and 2D engineering/manufacturing drawings in AutoCAD.
- Worked on process piping, lubrication systems, seal piping, base frames, foundations, MTO/BOM and package integration.
- Led and trained a team of 6 designers/engineers: assigned design/drafting work, reviewed deliverables and drawings, and guided junior members.
- Supported customer/vendor coordination, manufacturing technical queries and FAT activities.

CORE EXPERTISE

Compressor Packages	PSA Nitrogen
Fuel Systems	Storage Tanks
Mechanical Skids	Package Integration
Process Piping	Lubrication Systems
Seal Systems	3D Mechanical Design
Manufacturing Drawings	MTO / BOM
FAT / SAT	

SOFTWARE

SolidWorks	AutoCAD
Autodesk Inventor	AutoCAD 3D

LEADERSHIP & COORDINATION

- Led and trained a team of 6 designers/engineers on Atlas Copco package work.
- Head of Design Department, Project Coordinator and Lead Mechanical Designer for the Emarat 40 FT MFU at Meta Green UAE.
- Assigned modelling and drawing work and reviewed junior deliverables.
- Trained junior engineers/designers.
- Coordinated with senior management, project managers, procurement, production, instrumentation and customers/vendors.

APPLICABLE STANDARDS & CODES

Project experience includes work associated with API 614, API 617, API 672, API 692, ASME B31.3, ASME Section VIII, UL 142, UL 2085, NFPA, ATEX/IEC, ADR where applicable, and ISO 8573-1.

Mar 2022 — Mar 2023

Design Engineer — Compressor Packages

DesignTech Systems · Atlas Copco offsite

- Compressor package 3D design, GA layouts and manufacturing drawings.
- BOM/MTO, interfaces and package integration; equipment sizing and calculations.
- P&ID / equipment-selection coordination, vendor/customer coordination and FAT/SAT support.

Dec 2021 — Feb 2022

Design Engineer — Apprentice / Trainee

DesignTech Systems

Mechanical design support for industrial equipment, silencers, air-exhaust systems and fabricated structures using Autodesk Inventor.

Oct 2020 — Nov 2021

Junior Design Engineer

Sai Jyothi Engineering Works

Mechanical/piping design and assembly support, 3D modelling and detailed drawings with GD&T, and fabrication/inspection support.

EDUCATION & LANGUAGES

B.Tech, Mechanical Engineering

Pragati Engineering College, JNTU
Kakinada · 2016–2020

Industrial Training

45-day training, Vizag Steel Plant — Blast
Furnace Operations

English · Hindi · Telugu

SELECTED PROJECT EXPERIENCE

C-01 · NYK N₂ Componder Package

4-stage N₂ compander · Korea · 29,605 kg/h. Complete 3D package, lubrication system, seal-gas system, base frame, piping, GA, MTO/BOQ, customer/vendor coordination, manufacturing support and FAT.

C-02 · Xiexing Rudong BOG Compressor

1-stage LNG boil-off gas compressor · China · 28,296 kg/h · −158.3 °C inlet. Complete 3D SolidWorks model, AutoCAD layout/manufacturing/foundation drawings, lube-oil system, seal piping and instrumentation integration; reviewed the analysis team's stress report; manufacturing/FAT support.

C-03 · Banyan FGB Natural Gas Compressor

4-stage natural-gas compressor · Singapore · 36,980 kg/h. Complete package redesign and integration including process piping calculations, lubrication, seal system, panels, base skid, customer tie-ins and manufacturing/FAT support.

C-04 · Sichuan H₂O₂ Compressor & Expander Package

2 compressors + 1 expander · China. Common package arrangement, process piping, shared lubrication integration, supports, lube-skid stress calculations, equipment/foundation load calculations for civil design, customer review, manufacturing support and FAT.

C-05 · Maoming Air Compressor

3-stage steam-turbine-driven air compressor · China · 95,919 kg/h · API 672. Complete 3D compressor package, plant layout, process piping, lubrication system, intercoolers and rundown tank; compressor-side coupling interface only (turbine customer-supplied); manufacturing/FAT support.

P-01 · ONGC PSA Nitrogen Generator

~160 Nm³/h N₂ · 99.9% purity · ~6 barg delivery. Main mechanical/package designer covering 3D design, equipment integration, calculations, MTO/BOM, multidisciplinary coordination, customer/vendor coordination, FAT and some SAT.

F-01 · Emarat Mobile Fueling Unit

40 FT dual-compartment MFU-3A · 42,600 L gross tank volume · UAE. Head of Design Department + Project Coordinator + Lead Mechanical/Package Designer from tender through design, procurement, manufacturing, FAT and SAT.